

Benjamin Dodge

dodgebc.github.io

bendodge@stanford.edu

ORCID: 0000-0002-2519-2219

EDUCATION

Ph.D. Candidate in Physics, Stanford University

2023 – present

Member of Kavli Institute for Particle Astrophysics and Cosmology

Advised by Professor Susan Clark

Bachelor of Arts in Physics, High Honors, Princeton University

2018 – 2022

Thesis: *Stellar Disk Tilting in Galaxy Mergers*, advised by Professor Mariangela Lisanti

PUBLICATIONS

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|--|---|
| [1] The Via Collaboration (including Benjamin Dodge)
<i>The Via Project: Overview of the Science, Instrument, and Survey</i> | arxiv:2606.18332 |
| [2] Benjamin Dodge , Philipp Frank, Susan E. Clark
<i>GraphGP: Scalable Gaussian Processes with Vecchia's Approximation</i> | PAI 2026
arxiv:2606.11402 |
| [3] Benjamin C. Dodge , Oren Slone, Mariangela Lisanti, Tim Cohen
<i>Dynamics of Stellar Disc Tilting from Satellite Mergers</i> | MNRAS 518 no. 2, 2870
arxiv:2207.02861 |
| [4] Dan Friedman, Ben Dodge , Danqi Chen
<i>Single-dataset Experts for Multi-dataset Question Answering</i> | EMNLP 2021
arxiv:2109.13880 |

PRESENTATIONS

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|--|------|
| Talk: AstroAI lunch talk, Center for Astrophysics, Cambridge, MA
<i>GraphGP: Scalable Gaussian Processes with Vecchia's Approximation</i> | 2026 |
| Poster: Stanford Conference on Physics and AI, Stanford, CA
<i>GraphGP: Scalable Gaussian Processes with Vecchia's Approximation</i> | 2026 |
| Talk: AI+Science: Accelerating Discovery, Stanford, CA
<i>GraphGP: Scalable Gaussian Processes with Vecchia's Approximation</i> | 2026 |
| Talk: MMT Symposium, Tucson, AZ
<i>Kinetic Tomography with Sodium Absorption in the Plane</i> | 2026 |
| Talk: Center for Decoding the Universe Winter Forum, Stanford, CA
<i>GraphGP: Scalable Gaussian Processes with Vecchia's Approximation</i> | 2026 |
| Poster: Stanford Data Science Conference, Stanford, CA
<i>Mapping Cold Halo Gas in 3d with the Via Survey</i> | 2025 |

SERVICE & OUTREACH

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| Member, KIPAC Tea Committee | 2026 – present |
| Host and Dome Operator, KIPAC Public Stargazing Nights | 2024 – present |
| Graduate Mentor, Stanford University Physics Society Mentorship Program | 2025 – present |
| Organizer, Center for Decoding the Universe Journal Club | 2025 – 2026 |
| Graduate Student Organizer, Stanford Physics Undergraduate Directed Reading Program | 2025 |
| Cosmic Magnetism Booth Volunteer, KIPAC Community Day | 2025, 2024 |

Organizing Committee, Stanford Physics Identity and Equity Virtual Program	2024
Organizing Committee, Stanford Physics Identity and Equity Summer Workshop	2024
Undergraduate Recruitment Working Group, Princeton Physics Equity, Diversity, Inclusion	2020 – 2022

TEACHING

Teaching Assistant, Stanford University

- *PHY113: Computational Physics* 2026
- *PHY100: Observational Astrophysics* 2025
- *PHY100/300: Observational Astrophysics* 2024

Undergraduate Teaching Roles, Princeton University

2020 – 2022

- Study session organizer for *PHY104: Electricity and Magnetism* (Spring 2022)
- Department tutor for *PHY106: Electricity and Magnetism* (Spring 2022)
- Department tutor for *PHY205: Classical Mechanics* (Fall 2021)
- Undergraduate Course Assistant for *ORF309: Probability and Stochastic Systems* (Spring 2021)
- Grader for *COS484: Natural Language Processing* (Spring 2021)
- Grader for *COS324: Introduction to Machine Learning* (Fall 2020)

Leader Trainer & Instructor, Princeton University Outdoor Action

2019 – 2022

- Lead week-long backpacking training trips for new orientation leaders
- Teach leadership, technical skills, and CPR certification classes

English Teaching Assistant, Fulbright Taiwan

2022 – 2023

- Co-teach 16 periods of English per week at ZhuangWei and GuoLing elementary schools in Yilan County

AWARDS

- NSF Graduate Research Fellowship 2023
- Fulbright Taiwan English Teaching Assistantship Award 2022 – 2023
- Allen G. Shenstone Prize in Physics and Departmental Service, Princeton University 2021
- Manfred Pyka Memorial Prize in Physics, two-time awardee, Princeton University 2019, 2020
- Speaker at Mary W. George Freshman Research Conference, *Can Computers Learn from Copyrighted Content?* 2019
- 1st place in Abstraction, Princeton University International Eye Photo Contest 2019